

The First-Order Axiomatisability of $\mathbf{CompHaus}^{\text{op}}$: An Already-Known Negative Answer

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Status: literature already answered. Lieberman, Rosický, and Vasey give a negative answer, and prove the stronger statement that $\mathbf{CompHaus}^{\text{op}}$ is not an abstract elementary category.

Source question

Marra and Reggio’s *Stone duality above dimension zero* asks in Remark 1.5 on page 3 of the source PDF:

“To the best of our knowledge it is an open problem whether $\mathbf{CompHaus}^{\text{op}}$ is axiomatisable by a first-order theory.”

Here $\mathbf{CompHaus}$ is the category of compact Hausdorff spaces and continuous maps [1].

Identification of the answer

Lieberman, Rosický, and Vasey explicitly state in the introduction of their later paper that their non-elementarity theorem answers Marra–Reggio’s Question 1.5 [2, p. 2]. Their Theorem 22 proves that the category of commutative unital C^* -algebras with unit-preserving $*$ -homomorphisms is not an abstract elementary category. Gelfand duality identifies that category with $\mathbf{CompHaus}^{\text{op}}$, so their Corollary 24 states, on page 8 of the supporting PDF,

$\mathbf{CompHaus}^{\text{op}}$ is not an abstract elementary category.

A category of models of a finitary first-order theory, with all structure homomorphisms, is an abstract elementary category in the supporting paper’s framework. Corollary 24 therefore gives the requested negative answer: $\mathbf{CompHaus}^{\text{op}}$ is not axiomatisable by a first-order theory. In fact, the failure of abstract elementarity is strictly stronger.

Provenance and scope

This is a literature identification, not a new mathematical result of the run. The supporting authors explicitly knew they were answering the source question: their introduction cites Marra–Reggio, Question 1.5. Exact-title, exact-question, author, and categorical-term searches located the later paper and a 2020 independent citation confirming the implication.

The answer concerns the first-order question in Remark 1.5. It does not negate Marra–Reggio’s main positive theorem, which supplies a finite equational axiomatisation using infinitary operations.

References

- [1] V. Marra and L. Reggio, *Stone duality above dimension zero: Axiomatising the algebraic theory of $C(X)$* , Adv. Math. **307** (2017), 253–287; arXiv:1508.07750.
- [2] M. Lieberman, J. Rosický, and S. Vasey, *Hilbert spaces and C^* -algebras are not finitely concrete*, J. Pure Appl. Algebra **227** (2023), no. 4, 107245; arXiv:1908.10200; doi:10.1016/j.jpaa.2022.107245.